

PORTFOLIO DECISION SYSTEM (PDS)

Architecture, Governance, and Operating Framework

A framework for combining independent strategy targets into one monitored portfolio

SYSTEM DOCUMENTATION v1.4

This document describes the current three-layer PDS architecture - PDS Core, Reinforcement-Learning Adaptive Risk Control, and Dynamic FX - along with the rules for decision timing, history, accounting, monitoring, and public reporting.

SlackQuant Research

Version 1.4 | 26 September 2026

research.slackquant.com/systems/pds/

Document Summary

Field	Details
Document	Portfolio Decision System (PDS): Architecture, Governance, and Operating Framework
Version	1.4
Release date	2026-09-26
System type	Portfolio Decision & Operating System
Public scope	Public view of the current Core, Adaptive, Preview, FX, Performance, and Portfolio information. Environment-specific infrastructure, brokerage-account data, and protected implementation details are excluded.
Current architecture	Layer 1: PDS Core (F2R 10% + ADAA 90% at the current monthly top-level target). Layer 2: PDS Adaptive - Reinforcement-Learning Adaptive Risk Control. Layer 3: Dynamic FX - monthly USD/KRW implementation overlay.
Decision and rebalance cycle	Monthly signals; rebalance at the first U.S. trading-day close available to both strategies after the signal month ends.
Performance measurement	Daily returns are the performance source of record. Monthly and annual figures are derived from the continuous daily series, using 252 trading days for annualization.
Disclosure policy	Public and local dashboards present the same reader-facing operating information. The public version excludes local paths, runtimes, caches, credentials, repository and debug infrastructure, brokerage-account information, and protected controller artifacts.

How this document is organized

Sections 1-4 explain what PDS does and how the three operating layers fit together. Sections 5-9 cover construction, timing, performance, portfolio status, Dynamic FX, and governance. Sections 10-12 cover monitoring, operations, and public disclosure. Sections 13-15 cover release validation, related methods, and limitations. The appendices summarize decision status, public resources, and the v1.4 specification.

OVERVIEW

1. Executive Overview

Portfolio Decision System (PDS) combines target portfolios from independently developed investment strategies into a single portfolio process. It leaves each strategy's investment logic unchanged and standardizes only the portfolio-level rules for integration, timing, accounting, monitoring, and disclosure. The underlying strategy outputs remain separately identifiable.

PDS has three operating layers. PDS Core combines Forecast-to-Rank Allocation (F2R) and ADAA under the current strategic allocation. PDS Adaptive starts from Core and adds Reinforcement-Learning Adaptive Risk Control, a bounded risk-control layer rather than a standalone alpha or security-selection model. Dynamic FX is a separate currency layer that applies the same USD/KRW rule, execution clock, and cost schedule to the relevant Core or Adaptive USD return path. Keeping the layers separate makes attribution and decision responsibility clear.



A useful distinction

F2R and ADAA source targets, the Core Official Decision, Adaptive decisions, Dynamic FX decisions, Preview, Forward Shadow, standalone provider records, and reconstructed PDS histories do not carry the same status. Core defines the base portfolio; Adaptive changes only its own risk budget; Dynamic FX changes only the currency layer.

What PDS does

- Combines strategy targets into one portfolio using a consistent set of portfolio rules.
- Applies the same rebalance, turnover, cost, and performance rules when the strategies are evaluated together.
- Keeps the current Official Decision separate from Preview and Forward Shadow outputs.
- Retains the last valid Official Decision when required data or timing checks do not align.
- Publishes Core, Adaptive, Preview, FX, Performance, and Portfolio information after the release checks pass.
- Runs PDS Core and PDS Adaptive in parallel so Core remains unchanged while Adaptive risk control is evaluated and operated separately.

What PDS does not do

- It does not redesign the investment logic of F2R or ADAA.
- It does not treat Preview, Forward Shadow, or FX research as an approved portfolio decision.
- An Official Decision records the PDS target; it is not a broker fill or an account-level trade confirmation.
- The public PDS dashboard presents the same reader-facing operating information as the local dashboard while excluding local paths, runtimes, caches, credentials, repository controls, and debug tooling.
- PDS Adaptive does not overwrite or backfill PDS Core history, and an Adaptive failure does not change the Core portfolio decision.
- RL does not select securities or replace Core. It adjusts the Adaptive risk budget within the approved bounds.

ARCHITECTURE

2. System Architecture

PDS has three layers. Approved ADAA and F2R targets form Core. Adaptive applies a separately controlled Reinforcement-Learning risk-control layer to Core. Dynamic FX adds a monthly currency overlay to the Core and Adaptive USD paths. The layers share timing and accounting rules where needed, while their decisions and historical records remain separate.

Stage	Step	What happens
01	Strategy targets	Each strategy produces a target portfolio according to its own rules.
02	Common format	Targets are converted to a consistent asset-and-weight format so they can be combined without changing the underlying strategy rules.
03	Input checks	Strategy identity, date, timing, completeness, and weight consistency are checked before use.
04	Strategy review	Research evidence and practical portfolio value are reviewed before a strategy is approved for a portfolio role.
05	Portfolio combination	Approved targets are combined using the current PDS allocation policy.
06	Portfolio decision	The monthly PDS Core target is set under the common decision and execution clock. PDS Adaptive may apply its own risk budget, while Dynamic FX makes a separate monthly currency decision without changing the underlying USD target.
07	Monitoring and review	Core, Adaptive, Dynamic FX, Current MTD, Preview, Forward Shadow where applicable, and completed performance are monitored under defined clocks, with each view retaining its own status.

What PDS standardizes

PDS never recreates a missing strategy output. It uses only source targets that can be traced to the underlying strategy and pass the required date, timing, and completeness checks. If a required input cannot be aligned, the update is not adopted and the last valid Official Decision remains in place.

Independent strategies, common portfolio rules

F2R and ADAA retain their own investment logic. PDS standardizes only the elements that must be common at portfolio level: target format, timing, portfolio combination, performance measurement, and monitoring.

How the components fit together

System	Role in PDS	What remains separate
F2R	Produces a machine-learning cross-asset target portfolio used by PDS.	F2R retains its forecasting, ranking, and target-construction rules.
ADAA	Produces a multi-asset target portfolio used by PDS.	ADAA retains its allocation rules and target construction.
PDS	Combines the strategy targets and applies the common portfolio rules.	PDS does not recreate or override the underlying strategy models.
PDS Adaptive	Applies Reinforcement-Learning Adaptive Risk Control as a bounded layer on top of PDS Core.	It is not a standalone alpha or full portfolio-selection engine. The current state may be public, while exact controller parameters and training artifacts remain internal.
Dynamic FX	Applies the monthly USD/KRW overlay to Core or Adaptive USD paths.	The currency layer is separate from the underlying portfolio decision; spot-based monitoring is not broker-level forward/NDF execution evidence.

STRATEGY INPUTS

3. Strategy Inputs and Responsibilities

F2R and ADAA remain responsible for their own calculations. PDS takes their dated target portfolios, verifies the strategy identity, dates, timing, and weights, and combines them under PDS rules. A missing strategy output is never recreated inside PDS.

Checks before a target is used

- The correct strategy, target date, and relevant version are identified.
- Signal month, holding month, market-data cutoff, and rebalance timing line up.
- Weights are complete, properly scaled, and traceable to the strategy output.
- Strategy-level costs and PDS rebalancing costs are kept separate so the same cost is not counted twice.
- Missing, stale, or conflicting outputs are not replaced by estimates; the last valid Official Decision remains in place.

Standalone strategy history and PDS reconstructed history

F2R and ADAA retain their own standalone completed-performance records: F2R from 1 May 2017 onward under its current model, and ADAA from 2 June 2008 onward under its standalone strategy record. PDS also maintains a separate retrospective current-definition reconstruction for cross-system comparison, applying the current provider definitions and PDS timing and accounting rules across the available historical period. This longer PDS series is a research and comparison record; it does not replace either standalone record or imply that the current PDS policy was operated historically. Adaptive fixed-policy comparison history is likewise kept separate from its operational decision ledger.

Record	Purpose	How it is used
Standalone strategy record	Preserves each provider's own current-model or current-strategy completed-performance record, including its native timing and cost conventions.	F2R standalone completed performance starts 1 May 2017; ADAA standalone performance starts 2 June 2008. These records are not replaced by PDS reconstructions.

Record	Purpose	How it is used
PDS current-definition reconstruction	Applies the current provider definitions and PDS portfolio rules retrospectively over the available historical period for cross-system comparison.	The Core / Core + Dynamic FX common reader period is 2 December 2005 through 31 August 2026 for this release. It is research/comparison history, not a historical live-policy claim.
Public dashboard	Publishes the current reader-facing operating information after the release checks and removal of environment-specific details.	Used to show the current monitored state and completed performance without exposing environment-specific infrastructure.
Adaptive operational history	Post-deployment decisions and realized accounting only.	Separate from fixed-policy reconstructed comparison history and PDS Core; no retrospective splice or backfill.

Historical and operational records are kept separate
Standalone F2R and ADAA records, the retrospective PDS current-definition reconstruction, Adaptive fixed-policy comparison history, Adaptive operational decisions, and the public operating mirror remain separately identifiable. Publication does not turn one series into another.

CORE PORTFOLIO

4. Current Core Construction

PDS Core is the base portfolio. Its current monthly top-level target is F2R 10% and ADAA 90%. Each strategy determines its own asset weights; PDS combines the source targets under the common monthly execution, turnover, cost, and daily-accounting rules. The SlackQuant landing page emphasizes the architecture rather than the exact ratio; the dashboard and this document provide the construction detail.

PDS Core monthly target = F2R 10% + ADAA 90%

Total target weight = 100%

The ratio is part of the detailed system construction; the SlackQuant landing page does not present it as the identity of the system.

The provider allocation is set at each monthly rebalance, not reset every day. After rebalancing, both provider weights and underlying asset weights drift with market returns until the next scheduled rebalance.

PDS Adaptive

PDS Adaptive is a parallel Core-based variant driven by an approved frozen RL policy. In a NORMAL state, the risk budget is 100% of Core. In a DEFENSIVE state, the policy can choose a lower risk budget within the approved bounds: Core asset weights are scaled by that budget and the residual is held in BIL. DEFENSIVE therefore identifies a state rather than a single fixed cash allocation, so different DEFENSIVE months can use different risk budgets. Historical Adaptive targets are reconstructed from the fixed policy for comparison; they are not presented as a retroactive live-policy record. The dashboard may show the current Official and

Preview state, risk budget, and portfolio, while training artifacts, model seeds, thresholds, and exact controller parameters remain internal.

Target and portfolio weights

Measure	Meaning	Use
Official target	Portfolio weights set for the next holding period at the monthly rebalance.	Reference for the scheduled rebalance.
Post-rebalance weights	Portfolio weights immediately after the scheduled rebalance.	Starting weights for the next holding period.
Pre-rebalance weights	Portfolio weights immediately before the next rebalance after market movements.	Used to calculate turnover and rebalancing costs.
Current Mark	Provisional valuation of the incomplete holding period.	Monitoring only; does not change completed history or the Official Decision.
Execution-day opening exposure	The drifted portfolio held from the prior close into the scheduled execution close.	Earns the execution-day return before the new target is applied; this can differ between Core and Adaptive even when the new close target is identical.

Why weight drift matters

Resetting provider weights to their strategic targets every day would define a different strategy, with different exposures, turnover, and returns. PDS therefore lets weights drift between monthly rebalances and restores the target allocation only at the next scheduled rebalance.

Key point

The provider allocation is set at the monthly rebalance and then allowed to drift; it is not reset to constant weights each day.

TIMING

5. Decision and Rebalance Timing

PDS uses a monthly decision cycle. After the signal month closes, the next target is determined from information available at that cutoff. Rebalancing occurs at the first U.S. trading-day close available to both F2R and ADAA. The inherited portfolio earns the close-to-close return through that execution date; transaction cost is booked at the close; the new target starts with the next return interval. The same rule applies to Core and Adaptive. As a result, an Adaptive risk budget of 100% can still produce a different execution-day return if Adaptive entered the day with a different inherited book.



How the monthly cycle works

- The completed signal month determines the target for the next holding period.
- The portfolio is not rebalanced at the same month-end close that produced the signal.
- The existing portfolio remains in place through the rebalance close.
- The new weights apply from the next close-to-close return interval.
- Monthly performance is labeled by the holding month in which the return is realized.
- Calendar-year performance is assigned to the year in which the holding-period returns occur.

When dates do not line up

Strategy targets are combined only when their signal dates and relevant market-data dates are aligned. If a required strategy output is missing or not yet available, PDS does not estimate it. The last valid Official Decision remains in force until the required inputs are available and aligned.

Why timing matters

The signal cutoff, execution close, and first return interval of the new target are different points in time. A 100% Adaptive risk budget set at month-end means the target will match Core after execution; it does not replace the inherited Adaptive book that earns the execution-day return. This is part of the portfolio definition, not a reporting convention.

PERFORMANCE MEASUREMENT

6. Performance Measurement and Transaction Costs

All PDS performance statistics are derived from continuous daily returns. Monthly, year-to-date, annual, cumulative, drawdown, volatility, Sharpe, and Calmar figures use this daily series, with 252 trading days for annualization. Completed Performance ends at the latest completed holding month; Current MTD is provisional and shown separately. Core and Adaptive follow the same convention, but they can differ on an execution day because that day's return is earned by the pre-trade book before the new close target is applied.

$$\begin{aligned}\text{Daily wealth: } W_d &= W_{(d-1)} \times (1 + r_{d,\text{net}}) \\ \text{Monthly return: } R_{\text{month}} &= W_{\text{month-end}} / W_{\text{prior-month-end}} - 1\end{aligned}$$

Return and risk calculations

Daily returns are the base frequency. Annualized volatility and Sharpe use 252 trading days, and drawdown is measured from the running peak of daily wealth.

Turnover and rebalancing cost

Turnover is calculated from the portfolio weights immediately before the rebalance and the new target weights. PDS defines one-way turnover as one half of the sum of the absolute weight changes.

$$\begin{aligned}\text{One-way turnover} &= 0.5 \times \sum_i | \text{target weight}_i - \text{pre-rebalance weight}_i | \\ \text{PDS rebalancing cost} &= 2 \times \text{one-way turnover} \times 0.001\end{aligned}$$

The current cost assumption is 10 basis points per side. A full portfolio replacement (100% one-way turnover) therefore carries a 20-basis-point round-trip cost.

Avoiding double counting

Costs already reflected in the F2R or ADAA return series are not charged again at PDS level. PDS adds only the incremental cost of moving from the drifted pre-trade book to the new PDS target. Adaptive follows the same rule and rebalances directly from its own inherited book; it does not first pay for a hypothetical Core rebalance. On an execution date, the old book earns that day's return, turnover and cost are booked at the close, and the new target applies from the next return interval.

Consistency check

All published PDS return figures follow the same timing, turnover, cost, and daily performance rules. Any unexplained difference between views is resolved before release.

PORTFOLIO STATUS

7. Official Decision and Monitoring Views

The dashboard distinguishes portfolio decisions from monitoring views. The Core Official Decision is the current Core target, while PDS Adaptive runs in parallel with its own state and risk budget. Current Mark, Preview, and Forward Shadow remain monitoring views. The public dashboard can show current Core and Adaptive state, portfolio detail, Preview, performance, and Dynamic FX information under the same timing rules as the local dashboard.

View	What it shows	How it is used
Core Official Decision	The target portfolio in effect for the current holding period.	Used as the PDS portfolio target. It is not a broker fill or trade confirmation.
Current Mark	A provisional view of performance for the current, incomplete holding period.	Monitoring only; it does not change the Official Decision or completed performance history.
Preview	What the next target would look like if the current month ended using the latest available data.	Monitoring only; it is not executed and no Preview performance is reported.
Forward Shadow	A separately tracked alternative to the Active Core used for comparison.	Not traded as part of the Active Core; any performance is kept separate.
Public Operating Dashboard	Public reader-facing view of the current operating state.	Shows current monitoring information and completed performance under the same clocks as the local dashboard.
PDS Adaptive	Parallel Core-based variant with NORMAL or DEFENSIVE risk-control state.	Current state, risk budget, holdings, Preview, and performance may be shown publicly; low-level controller internals remain protected.

More recent does not mean official

A newer Preview or Forward Shadow result does not replace the Official Decision. It simply adds more recent monitoring information.

Changing portfolio status

A monitoring view becomes part of Core only through the regular monthly decision process. PDS Adaptive changes only its own parallel portfolio through the risk-control layer. A newer timestamp does not change the status of either view.

KRW INVESTOR CONTEXT

8. Currency Exposure and the FX Overlay

For a KRW-based investor holding U.S.-listed multi-asset ETFs, exchange-rate moves can materially affect portfolio returns and risk. PDS therefore manages currency exposure at the total-portfolio level rather than embedding a separate FX decision inside one underlying strategy.

FX implementation and clocks

FX view	Status	How to interpret it
Dynamic FX operating layer	Current monitored currency layer	Shows the current Official and Preview monthly hedge state, Current MTD, and completed Core/Adaptive Dynamic-FX performance under the defined spot-sensitivity rule and cost schedule.
Broker-level forward/NDF realization	Not represented by the current spot-based proxy	Realized hedge P&L would require tradable forward/NDF data, actual availability, carry/basis, contract-roll rules, execution, and realized transaction costs.
FX clock terminology	Three distinct timestamps	Latest raw FX observation = newest source-feed timestamp; Latest admissible FX observation = newest FX point allowed by the PDS/common calendar and usable for z-score/hedge decisions; Portfolio/Preview market-through = latest date through which the relevant portfolio/Preview market inputs are complete. These dates need not be identical.

Dynamic FX is the monthly USD/KRW implementation layer for both PDS Core and PDS Adaptive. The same FX rule, execution clock, and cost schedule apply to both; only the underlying USD portfolio path may differ. The public dashboard can show the current Official FX decision, Preview state, Current MTD, and completed FX performance. These spot-based results are an implementation proxy, not realized broker-level forward/NDF hedge P&L.

Why the distinction matters

- Spot-FX sensitivity is not the same as realized forward/NDF hedging performance.
- Forward carry, basis, contract rolls, and changes in hedge ratios can affect actual hedging results.
- An FX series developed in a strategy research project is not automatically adopted by PDS.
- A historical FX series should be interpreted using its stated signal, admissibility, and market-through clocks rather than treated as a substitute for an executed hedge record.

GOVERNANCE

9. Research Evidence, Portfolio Usefulness, and Adoption

PDS separates three questions: Does the evidence support the research claim? Does the strategy add value in the portfolio role being tested? Has it been adopted into the Official portfolio? A positive answer to one does not imply a positive answer to the others.

Review	Question	What it determines
Research review (Gate R)	Does the evidence support the research claim being made?	Whether the research claim passes, fails, or remains inconclusive.
Portfolio-role review (Gate P)	Does the strategy add enough practical value for the portfolio role being tested?	Whether the strategy is suitable for that role, unsuitable, or still inconclusive.
Official adoption	Has the strategy or allocation actually been approved for the PDS portfolio?	Only an explicit adoption decision changes the Official portfolio.

These are separate decisions

A strategy can have sound research evidence without being useful for the role being considered. It can also look useful in a portfolio test without supporting a new research claim. Neither result changes the Official Core unless the strategy is explicitly adopted.

When a review does not pass

If required data, timing, or evidence cannot be verified, the proposed update is not used. A Preview issue does not alter the last valid Core Official Decision. A strategy stays outside the Official Core until a later review supports adoption. Adaptive follows the same fail-closed principle: if the frozen policy, state inputs, or integrity checks fail, it produces no new decision while Core continues independently.

MONITORING

10. Monitoring Views

PDS monitors the current Core Official portfolio, PDS Adaptive, Preview, Current MTD, Dynamic FX, completed performance, and portfolio history. Forward Shadow, where shown, remains a separate comparison view. Each view has a defined role and clock so monitoring information is not mistaken for a portfolio decision. After validation, the public dashboard shows the same operating information while excluding environment-specific infrastructure and protected implementation details.

Dashboard views

Dashboard view	What it shows	How to interpret it
Core	Current Official decision, Current MTD, provider contributions, and the primary portfolio state.	The Official target remains authoritative; Current MTD is provisional and separate from completed performance.
Adaptive	Current state, risk budget, target relation to Core, holdings, and risk-control history.	Parallel to Core. NORMAL uses 100% risk budget; DEFENSIVE can use a lower risk budget, with the residual held in BIL.
Preview	Core Preview and the separately evaluated Adaptive prospective state for the next holding month.	Target-only monitoring; not executed and excluded from completed performance.
FX	Current Official and Preview FX state, admissible observation clock, Current MTD, hedge history, and completed performance.	The same FX rule and cost schedule apply to Core and Adaptive USD paths; spot-based results are not broker-level NDF execution evidence.
Performance	Core, Core + Dynamic FX, Adaptive, Adaptive + Dynamic FX, and provider/reference comparisons with stated performance periods.	Completed history remains separate from provisional Current MTD. Reconstructed PDS history and standalone provider records are labeled separately.
Portfolio	Current-definition Core, Adaptive, and monitoring target histories in fixed ETF columns, with downloadable Excel.	Preview and Official rows are labeled explicitly. Adaptive history also shows State and Risk Budget so DEFENSIVE intensity is visible.
Forward Shadow (where shown)	A separately tracked alternative to the Active Core used for comparison.	Comparison only; display or performance does not make it part of the Official Core.

When data are unavailable

If a strategy update, Preview input, timing reference, or Current Mark is unavailable, PDS keeps the last valid Official information and marks the newer item as unavailable. Missing data are not estimated merely to keep the dashboard populated.

OPERATIONS AND PUBLICATION

11. Operating and Publication Process

PDS decisions are produced in the controlled operating environment. Publication creates a public copy of the approved dashboard rather than a separate delayed or simplified investment product. It does not create a new portfolio or trading decision; it distributes information that has already passed the operating checks.



Main operating steps

Step	Purpose
Refresh and rebuild	Update strategy inputs and rebuild the PDS dashboard. If a monitoring input is unavailable, the last valid Official Decision remains unchanged.
Review the current state	Check the Official Decision, Current Mark, Preview, Forward Shadow, and related changes before any release is prepared.
Prepare the public update	Generate the public dashboard, remove environment-specific infrastructure, and verify that the public version preserves the approved operating information and clocks.
Approve and publish	Review the changes, confirm the published figures, and release the update to the SlackQuant site.

Publication checks

1. Check the PDS public figures, labels, links, and disclosure boundaries.
2. Build the public pages and inspect the rendered result.
3. Confirm that the downloadable and linked resources match the approved release.
4. For a new system-document version, review the rendered document page-by-page; fix and re-render each page with a finding before moving to the next page.
5. Publish only after the checks above pass.

Final review remains explicit

Automation can update data, run checks, and prepare a release, but a substantive public update still receives a human review before publication.

PUBLIC DISCLOSURE

12. Public Disclosure

The public PDS dashboard is the public reader-facing counterpart of the local dashboard. It can show current Core and Adaptive state, portfolio detail, Preview, Dynamic FX monitoring, and performance. The public/private boundary is mainly technical: local paths, runtimes, caches, credentials, repository controls, debug tooling, brokerage-account information, and protected model artifacts remain private.

Published	Internal / suppressed
System architecture, provider relationships, governance, and system documentation	Local filesystem paths, runtime/cache/repository locations, credentials/secrets, and debug/runner tooling
Current Core and Adaptive decision state, detailed portfolio targets, and operating clocks	Low-level Adaptive controller training artifacts, model seeds, exact thresholds/parameters, and implementation-specific internal identifiers
Current Mark, Preview, Forward Shadow where shown on the dashboard, and performance monitoring	Brokerage-account holdings, realized account P&L, order-routing infrastructure, and trade confirmations
Current Dynamic FX monitoring, Preview FX state, Current MTD, hedge history, and completed Dynamic-FX performance	Research branches and experimental artifacts that have not been promoted to the current system
Research, Quantitative Methods, release validation, and versioned system documentation	Machine-specific metadata that is not needed to interpret the published system
Core, Core + Dynamic FX, Adaptive, and Adaptive + Dynamic FX performance with explicit performance periods and historical-series labels	Protected security-sensitive implementation details not required to reproduce the published operating state

How public performance is dated

Monthly performance is labeled by the month in which the portfolio return is realized, not by the month in which the signal was formed. Annual figures follow the same holding-period convention.

Status of public materials

Public PDS pages are research and monitoring tools, not execution interfaces or trade confirmations. Their dates and status follow the same decision, admissibility, and market-through rules used by the operating dashboard.

Publication rule

A release is blocked if figures, clocks, labels, portfolio detail, or cross-system references do not agree, or if environment-specific or security-sensitive material appears in the public artifact. Publication does not change portfolio decision status.

RELEASE CHECKS

13. Release Validation

For v1.4, the current 10/90 Core construction, the separation of standalone and reconstructed histories, execution-close accounting, Adaptive state and risk-budget treatment, Dynamic FX clock terminology, dashboard terminology, and public documentation links were checked against the operating records. The table summarizes those release checks; it documents the system as released and is not a forecast.

Check	Result
Core allocation	The current monthly top-level Core target is F2R 10% / ADAA 90%. The ratio is documented here and in the detailed dashboard, while the platform landing page emphasizes the architecture.
Current-definition Core history	The displayed long history applies the current 10/90 Core definition consistently across the reconstructed period. Earlier top-level mixes remain internal research references and are not spliced into the reader-facing series.
Standalone provider history	F2R standalone completed performance remains its 1 May 2017 onward current-model record; ADAA standalone performance remains its 2 June 2008 onward strategy record. The longer PDS provider/reference histories are separately labeled retrospective current-definition reconstructions.
Rebalance timing	The monthly decision and execution-close dates matched the timing rules described in Section 5.
Transaction costs	Drift-based turnover and the 10 bp-per-side PDS rebalancing cost assumption matched the operating calculation; provider-level costs are not counted twice.
Performance measurement	Completed Performance is derived from the daily return series with 252-trading-day annualization; Current MTD remains provisional and separate.
Performance periods at release	Core and Core + Dynamic FX use the common reader period 2 December 2005 through 31 August 2026. Adaptive fixed-policy comparison uses 3 May 2021 through 31 August 2026. Standalone provider records retain their own dates.
Display consistency	Core, Adaptive, Preview, FX, Performance, and Portfolio views matched the current terminology and operating clocks.
Adaptive state / risk budget	NORMAL maps to a 100% Core risk budget. DEFENSIVE may use different lower risk budgets by month; Core weights are scaled by the risk budget and the residual is allocated to BIL. State and Risk Budget are shown together in Adaptive history.
Core non-mutation / failure isolation	PDS Core allocation and accounting remain unchanged by Adaptive. Adaptive failures are contained within the Adaptive branch and do not block or rewrite Core operation.
Public documentation wiring	The v1.4 PDF is the current System Documentation resource for publication; earlier public PDS documentation versions are retired from the current platform surface.

RELATED METHODS

14. Related Quantitative Methods

The SlackQuant Quantitative Methods library provides background on the techniques referenced in this document. The entries below are the methods most directly related to PDS.

ID	Method	Relevance to PDS
QM007	Portfolio Backtesting and Rebalancing	Explains monthly rebalancing, weight drift, and next-period return timing.
QM008	Drawdown, Underwater Duration, and Recovery	Covers drawdown depth, duration, recovery, and running peaks.
QM009	Turnover, Transaction Costs, and Net Performance	Explains turnover, transaction costs, and net portfolio returns.
QM011	Strategy Similarity and Decision Diversification	Provides the basis for comparing strategy decisions rather than return correlation alone.
QM013	Benchmark-Relative Portfolio Evaluation	Covers active return, tracking error, information ratio, relative wealth, and relative drawdown.
QM014	Information Timing, Data Vintages, and Real-Time Macro Data	Explains release timing, data vintages, revisions, and the information available at each decision date.
QM028	Reinforcement Learning for Adaptive Portfolio Risk Control	General background on RL-based risk control. The PDS-specific fixed-policy design, evidence boundary, execution clock, and accounting are defined in this document.
QM029	Dynamic Currency Hedging and FX Overlay Accounting	General background on FX overlays. The PDS-specific raw/admissible observation rules, market-through clock, monthly hedge decision, and Core/Adaptive treatment are defined here.

Methods library: research.slackquant.com/methods/

How these references are used

The Methods library explains the general techniques; this document specifies how PDS uses them. QM028 covers Reinforcement Learning for Adaptive Portfolio Risk Control, while this document defines the PDS fixed-policy design, clocks, evidence boundary, and accounting. QM029 covers Dynamic Currency Hedging and FX Overlay Accounting, while this document defines the PDS FX observation rules, monthly decision clock, and Core/Adaptive overlay treatment.

LIMITATIONS

15. Limitations and Revision Policy

Limitations

- Historical results, including observations from live operation, do not guarantee future returns.
- The current F2R/ADAA allocation is the present PDS Core policy. It is a system design choice, not a claim that the ratio is universally or permanently optimal.
- F2R and ADAA can still share market exposures even though their decision processes differ. Their combination should therefore be evaluated from the observed portfolio results rather than assumed to be diversified by design.
- Transaction costs are modeled assumptions. Actual trading costs can differ.
- Dynamic FX uses spot-FX sensitivity and an assumed hedge-renewal cost schedule. It should not be read as realized broker-level forward/NDF hedging performance.
- Preview and Forward Shadow are monitoring tools and do not change the Official portfolio unless they are separately adopted through the regular decision process.
- The Adaptive fixed-policy historical comparison comes from exploratory research and should not be read as independent confirmatory evidence. Operational Adaptive decisions and realized accounting are tracked separately.
- This document describes the PDS portfolio framework. Broker fills, custody records, account holdings, and trade confirmations are outside its scope.

When a new version is needed

Change	Update needed
Presentation-only edit	Normally no new version unless the meaning changes.
Routine data update or new completed month	No new version; the historical record is simply extended.
Core strategy membership or target allocation	New document version.
Rebalance timing, turnover, or transaction-cost treatment	New document version.
Definition or role of Official Decision, Preview, or Forward Shadow	New document version.
Public disclosure policy	New document version.
Dynamic FX role, implementation method, or accounting treatment	New document version.
PDS Adaptive role, policy binding, evidence boundary, execution semantics, or performance-accounting treatment	New document version.

Versioning rule

A new version is issued when portfolio construction, timing, cost treatment, decision status, or disclosure policy changes materially. Routine monthly data updates do not require a new version.

Appendix A. Decision and Publication Status

View	Current?	Can change Core target?	Performance?	Published?
Core Official Decision	Yes	Yes - through the regular monthly decision process	Realized under the execution-close accounting rule	Yes - current target and operating clocks are shown in the public dashboard
Current Mark	Yes - incomplete period	No	Provisional current-period view	Yes - monitoring status is explicit
Preview	Yes - incomplete period	No	No executed Preview performance	Yes - target-only/non-authoritative status is explicit
Forward Shadow	Yes - monitoring	No	Separate comparison performance may be tracked	Yes where shown on the public dashboard; display alone does not change its status
Public Dashboard	Current published snapshot	No - publication does not change portfolio decisions	Shows current monitoring information and completed performance	Yes
Dynamic FX	Yes - separate monthly currency layer	No - does not change the underlying Core target	Current MTD and completed overlay performance	Yes - current Official/Preview state and completed performance; broker-level NDF execution is not implied
PDS Adaptive	Yes - parallel operating variant	Changes only the Adaptive target through its risk-control layer; never changes Core	Fixed-policy historical comparison plus separate post-deployment operational accounting	Yes - current state, risk budget, and portfolio can be shown; low-level controller internals remain protected

Reading the table

- The Core Official Decision changes the Core target through the regular monthly process. PDS Adaptive changes only its own parallel target through the risk-control layer. Preview, Forward Shadow, and Current Mark remain monitoring views. Publication does not turn a monitoring view into an Official Decision.

Appendix B. PDS Resources

Resource	Purpose	Location
PDS System page	Overview of the system, PDS Core / PDS Adaptive architecture, current Core providers, and related public resources.	research.slackquant.com/systems/pds/
PDS Dashboard	Public operating view for Core, Adaptive, Preview, FX, Performance, and Portfolio.	research.slackquant.com/systems/pds/dashboard/
PDS System Documentation	Detailed explanation of Core/Adaptive construction, timing, accounting, monitoring, governance, and disclosure.	Linked from the PDS System page
Quantitative Methods	Background on the quantitative methods referenced in this document.	research.slackquant.com/methods/
Research	Separate empirical papers and research records related to the strategies and methods.	research.slackquant.com/research/

Start with the System page for an overview. The Dashboard shows the public operating view for current Core, Adaptive, Preview, FX, Performance, and Portfolio information. This document and the Methods library provide the supporting system and methodology detail.

Appendix C. PDS v1.4 Summary

Item	PDS v1.4
System	Portfolio Decision System (PDS)
System type	Portfolio Decision & Operating System
Current Core providers	F2R and ADAA
PDS Core	Base portfolio with a monthly top-level target of F2R 10% + ADAA 90%. The platform landing page emphasizes the architecture rather than the ratio.
Rebalance frequency	Monthly
Between rebalances	Provider and asset weights drift with market returns; there is no daily reset to strategic provider targets.
Rebalance timing	First common U.S. trading-day close after the signal month. The inherited pre-trade book earns through that close; the new target starts from the next close-to-close interval.
Turnover measure	One-way turnover is half the sum of the absolute changes between pre-rebalance and target weights.
Transaction-cost assumption	10 basis points per side.
Performance calculation	One continuous daily return series; monthly and annual results are derived from it.
Preview	Monitoring only; not executed and no Preview performance is reported.
History separation	Standalone provider performance remains separate from the longer retrospective PDS current-definition reconstruction. The reconstructed series is for research/comparison and is not presented as historical live-policy operation.
Public disclosure	Public and local dashboards present the same reader-facing operating information. Environment and security infrastructure, along with protected implementation details, remain private.
Dynamic FX	Third operating layer: monthly USD/KRW overlay applied under the same rule, execution clock, and cost schedule to Core and Adaptive USD paths. Current Official/Preview state and completed performance may be displayed.
FX evidence boundary	Spot-based operating sensitivity with assumed renewal cost is not broker-level forward/NDF realized P&L. Raw FX observation, admissible FX observation, and portfolio/Preview market-through are distinct clocks.
PDS Adaptive	Parallel Core-based variant using Reinforcement-Learning Adaptive Risk Control; RL is bounded risk control, not a standalone alpha or security-selection engine.
Adaptive operating states	NORMAL or DEFENSIVE. NORMAL uses a 100% Core risk budget; DEFENSIVE can use a lower risk budget that varies by month, with residual allocation to BIL. Current state and risk budget may be shown publicly; controller recipe and training artifacts remain internal.

Item	PDS v1.4
Adaptive evidence boundary	Historical Adaptive comparison is a fixed-policy reconstruction on its stated evaluation period; current Official/Preview decisions and post-deployment operational accounting remain separate. No retrospective live-policy claim or splice.
Adaptive failure behavior	Fails closed within the Adaptive branch; PDS Core remains independent and continues to refresh.

About this version

Version 1.4 describes the PDS framework in effect on 26 September 2026. It updates the current Core to F2R 10% + ADAA 90%, separates standalone provider performance records from the longer retrospective PDS current-definition reconstruction, clarifies that DEFENSIVE Adaptive months can use different risk budgets with the residual allocated to BIL, and aligns the document with the current Core, Adaptive, Preview, FX, Performance, and Portfolio dashboard structure. Timing, turnover, cost, and public/private disclosure principles remain unchanged except where explicitly described.

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